Comprehensive Business Analysis Report: Olist Marketplace Performance & Customer Experience

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Dataset: Brazilian E-Commerce Public Dataset by Olist (Sep 2016 – Oct 2018)

Target Audience: Olist Operations & Customer Experience Leadership

1. Executive Summary

An in-depth analysis of approximately 100,000 historical transactions on the Olist marketplace was conducted to evaluate multi-dimensional operational performance across orders, products, payments, fulfillment, and customer satisfaction.

The core finding of this investigation is that **logistics execution and delivery timeline adherence serve as the single largest determinant of customer review scores on Olist**. While overall order volume and platform gross revenue demonstrated steady month-over-month expansion across the two-year period, customer satisfaction fluctuated significantly during operational peak periods.

Key findings reveal that 1-Star negative reviews exhibit a **27.17% late delivery rate**, whereas 5-Star positive reviews demonstrate an exceptionally low **2.98% late delivery rate**. Furthermore, heavy/bulky product categories (such as office_furniture) experience systemic logistics delays, leading to satisfaction scores far below platform baselines. Strategic interventions must prioritize regional fulfillment decentralization, specialized logistics partnerships for bulky goods, and dynamic delivery algorithm recalibration.

2. Problem Statement & Business Context

Olist operates as an e-commerce marketplace integrator in Brazil, enabling small and medium enterprises (SMEs) to sell products through major e-commerce channels under a single contract. Sellers handle item dispatch, while Olist manages marketplace integrations and logistics carrier partnerships.

As Olist scales its seller base and geographic coverage, platform leadership requires a consolidated, data-backed understanding of:

1. Long-term marketplace growth and revenue trends versus customer review metrics.
2. The operational drivers causing variance between 1-Star ratings and 5-Star ratings.

3. Performance discrepancies across product categories, seller locations, and payment structures.
4. Actionable operational priorities to protect customer lifetime value (CLV) and marketplace reputation.

3. Data Architecture & Data Quality Adjustments

The analysis integrated nine distinct CSV datasets covering 99,441 orders, 112,650 order items, 103,886 payment records, and 100,000 review responses. To ensure statistical integrity and prevent data bias, the following operational data characteristics were handled:

- **Geolocation Granularity Fix:** The raw geolocation lookup table contained multiple lat/long entries per zip code (~52 rows per prefix). To prevent artificial row multiplication during table joins, spatial coordinates were aggregated to the geolocation_zip_code_prefix mean level before merging with customer and seller locations.
- **Filtering Delivered Status:** Non-completed orders (canceled, unavailable, processing) lacked complete fulfillment timestamps. Operational delivery timing metrics were strictly evaluated on delivered orders (order_status == 'delivered'; $n = 96,478$).
- **Timestamp Normalization & Feature Engineering:** Standardized datetime features were extracted to compute accurate delivery variance relative to promised delivery dates:
  $$\text{Delivery Delay (Days)} = \text{order\_delivered\_customer\_date} - \text{order\_estimated\_delivery\_date}$$
  (Where positive values indicate late arrival and negative values indicate early arrival).

4. Key Analytical Findings

A. Marketplace Performance Over Time (Core Q1)

- **Order & Revenue Expansion:** Order volume and platform gross merchandise value (GMV) demonstrated consistent exponential growth from late 2016 through mid-2018.
- **Satisfaction Decoupling During Demand Spikes:** Order volume peaks (most notably during Black Friday / holiday promotional events) correlated with temporary drops in average review scores, highlighting logistics carrier capacity bottlenecks during high-volume periods.

B. Delivery Timing & Customer Satisfaction (Core Q2 & Q6)

- **Late Delivery Asymmetrical Impact:** Delivery timing is the primary root cause of low review scores. Late delivery rates jump nearly **9-fold** from 5-Star reviews (**2.98%**) to 1-Star reviews (**27.17%**).
- **Early Delivery Multiplier:** 5-Star ratings are heavily driven by under-promising and over-delivering. Orders receiving a 5-Star rating arrived an average of **12.7 days prior** to their promised estimated delivery date. Conversely, 1-Star rating orders averaged arrival only **5.3 days prior** (with a substantial portion arriving well after estimated dates).

C. Geographic Disparities & Freight Friction (Core Q3)

- **Seller Concentration:** Over 60% of active sellers are concentrated in São Paulo (SP) and southern Brazilian states.
- **Regional Logistics Lag:** Orders shipped to customer locations in North and Northeast regions (e.g., Bahia, Ceará, Amazonas) incur significantly longer transit times, higher freight-to-product cost ratios, and higher late delivery rates compared to intra-state Southeast transactions.

D. Category Level Vulnerabilities (Core Q4)

- **Top Performing Categories:** Standard consumer commodities with simple fulfillment workflows scored highest: books_general_interest (4.45 avg score, 512 orders), luggage_accessories (4.32 avg score), and stationery (4.19 avg score).
- **Bottom Performing Categories:** Bulky, heavy, and assembly-required categories suffered the lowest platform scores: office_furniture (3.49 avg score across 1,273 orders), bed_bath_table (3.89 avg score across 9,417 orders), and furniture_decor (3.90 avg score). Freight damage risk and carrier handling delays are secondary drivers for these low scores.

E. Payment Choices & Financial Behavior (Core Q5)

- **Installment Reliance:** Credit card transactions accounted for the highest sales volume, with customers utilizing higher installment counts (6 to 12 installments) for high-ticket items.
- **Experience Neutrality:** Payment method choice itself showed no direct negative impact on review ratings, proving that customer dissatisfaction stems almost entirely from fulfillment execution rather than payment processing.

5. Strategic Recommendations & Action Plan

Based on empirical evidence from the dataset, Olist leadership should prioritize the following three strategic pillars:

Pillar 1: Regional Fulfillment & Warehouse Decentralization

- **Problem:** Shipping all goods from Southeast sellers to North/Northeast customers creates multi-week transit times and high late-delivery risks.
- **Action:** Establish regional fulfillment centers (3PL hubs) in key Northeast nodes. Encourage high-volume sellers to pre-stock fast-moving inventory locally.

Pillar 2: Logistics SLA Overhaul for Heavy & Bulky Goods

- **Problem:** Categories like office_furniture systematically drag down platform customer satisfaction due to standard postal/carrier mis-handling.
- **Action:** Contract specialized heavy-goods freight carriers with dedicated white-glove delivery capabilities and mandatory transit-tracking updates for furniture sellers.

Pillar 3: Dynamic Estimated Delivery Date (EDD) Recalibration

- **Problem:** Missed delivery promises severely damage customer trust, causing 1-Star review spikes.
- **Action:** Update the machine learning algorithm that calculates promised estimated delivery dates at checkout. Integrate real-time carrier capacity data and seasonal buffer days during peak periods to ensure Olist consistently under-promises and over-delivers.